Lecture 32 (*May 12*)

Reading: Section 12.3
Viewing: none

Last time, we discussed random vectors and covariance matrices in the discrete case, but you probably won't be surprised to hear that the formalism can be extrapolated to continuous probability distributions. Generalizing (282), a function:

$$p(x_1, \ldots, x_n) : \mathbb{R}^n \to \mathbb{R}_{\geq 0} \tag{294}$$

is called the **joint probability distribution** of a random vector $\boldsymbol{X}$ such as (289). The covariance matrix of this random vector is given by (290), if one replaces the sum by an integral:

$$K = E\Big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T\Big] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(x_1, \ldots, x_n) \begin{bmatrix} x_1 - \mu_1 \\ \vdots \\ x_n - \mu_n \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ \vdots \\ x_n - \mu_n \end{bmatrix}^T dx_1 \ldots dx_n \tag{295}$$

where:

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(x_1, \ldots, x_n) \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} dx_1 \ldots dx_n \tag{296}$$

A rather simple case is a random vector $\boldsymbol{Y}$ consisting of n <u>independent</u> normal distributions $Y_1, \ldots, Y_n$, i.e. consider the following joint probability distribution:

$$p(y_1, \ldots, y_n) = \frac{e^{-\frac{(y_1-\nu_1)^2}{2\Sigma_1} - \cdots - \frac{(y_n-\nu_n)^2}{2\Sigma_n}}}{\sqrt{(2\pi)^n \Sigma_1 \ldots \Sigma_n}} \tag{297}$$

(we denote them by y's instead of x's to avoid confusion later on). As a consequence of the identity:

$$-\frac{(y_1-\nu_1)^2}{2\Sigma_1} - \cdots - \frac{(y_n-\nu_n)^2}{2\Sigma_n} = -\frac{1}{2} \begin{bmatrix} y_1 - \nu_1 \\ \vdots \\ y_n - \nu_n \end{bmatrix}^T D^{-1} \begin{bmatrix} y_1 - \nu_1 \\ \vdots \\ y_n - \nu_n \end{bmatrix}, \quad \text{where} \quad D = \begin{bmatrix} \Sigma_1 & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \Sigma_n \end{bmatrix}$$

we may write formula (295) as:

$$D = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\nu})^T D^{-1} (\boldsymbol{y}-\boldsymbol{\nu})}}{\sqrt{(2\pi)^n \det D}} (\boldsymbol{y} - \boldsymbol{\nu})(\boldsymbol{y} - \boldsymbol{\nu})^T d\boldsymbol{y} \tag{298}$$

where $\boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ and $d\boldsymbol{y} = dy_1 \ldots dy_n$. Indeed, this is because the off-diagonal terms of the covariance matrix are 0 due to independence, while the diagonal terms are the variances of $Y_1, \ldots, Y_n$.

Now let's consider a general vector $\boldsymbol{X}$ of normal distributions $X_1, \ldots, X_n$. The setup here is that we consider an arbitrary $\boldsymbol{\mu} \in \mathbb{R}^n$ and an arbitrary positive definite matrix S, and let:

$$p(x_1, \ldots, x_n) = \frac{e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T S^{-1} (\boldsymbol{x}-\boldsymbol{\mu})}}{\sqrt{(2\pi)^n \det S}} \tag{299}$$

Remark. *The reason why we require S to be positive definite is that the quantity:*

$$E = \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^T S^{-1}(\boldsymbol{x} - \boldsymbol{\mu})$$

can be interpreted as the kinetic energy of some physical system of particles with velocities $x_1, \ldots, x_n$ (the fact that we subtract the average $\boldsymbol{\mu}$ means that we choose our coordinates so that the center of mass of the system stands still). The expression e^{-E} is called the Boltzmann factor in physics. Its presence means that velocity vectors $(x_1, \ldots, x_n)$ with lower energy arise with higher probability and those with higher energy arise with lower probability.

In the situation at hand, the covariance matrix (295) can be written as:

$$K = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T S^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}}{\sqrt{(2\pi)^n \det S}} (\boldsymbol{x} - \boldsymbol{\mu})(\boldsymbol{x} - \boldsymbol{\mu})^T d\boldsymbol{x} \tag{300}$$

To summarize: the entries of the random vector $\boldsymbol{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$ whose covariance matrix is K are no longer independent. However, we can find appropriate linear combinations which are independent, by diagonalizing the matrices S and K. This is PCA for continuous probabilities, and it goes something like this. First diagonalize the positive definite matrix S that appears in (299):

$$S = QDQ^T$$

where Q is orthogonal and D is diagonal (with positive diagonal entries). Then if we perform the substitution $\boldsymbol{Y} = Q^T \boldsymbol{X} \Leftrightarrow \boldsymbol{X} = Q\boldsymbol{Y}$ (note that the mean of $\boldsymbol{Y}$ is $\boldsymbol{\nu} = Q^T \boldsymbol{\mu}$), then (300) becomes:

$$K = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\nu})^T D^{-1}(\boldsymbol{y}-\boldsymbol{\nu})}}{\sqrt{(2\pi)^n \det S}} Q(\boldsymbol{y} - \boldsymbol{\nu})(\boldsymbol{y} - \boldsymbol{\nu})^T Q^T d\boldsymbol{y} =$$

$$= Q \left[\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\nu})^T D^{-1}(\boldsymbol{y}-\boldsymbol{\nu})}}{\sqrt{(2\pi)^n \det D}} (\boldsymbol{y} - \boldsymbol{\nu})(\boldsymbol{y} - \boldsymbol{\nu})^T d\boldsymbol{y} \right] Q^T \overset{(298)}{=} QDQ^T = S$$

(we are using $\det S = \det D$ and the fact that the determinant of the orthogonal matrix Q is ± 1). Since the covariance matrix of the random vector $\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$ is diagonal, the random variables $Y_1, \ldots, Y_n$ are uncorrelated normal distributions. Hence under orthogonal linear transformations Q, random vectors transform as $\boldsymbol{X} = Q\boldsymbol{Y}$, and their covariance matrix transforms as $K = QDQ^T$.

The principles of least squares also find their place in probability. Suppose A is a matrix, that $b_1, \ldots, b_m$ are random variables with variance $\Sigma_1, \ldots, \Sigma_m$, and you are trying to minimize the error in the least squares approximation to the linear system:

$$A\boldsymbol{v} \sim \boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \tag{301}$$

The appropriate notion of error in the experiment is:

$$\varepsilon = \sum_{i=1}^{n} \frac{(\boldsymbol{b} - A\boldsymbol{v})_i^2}{\Sigma_i} \tag{302}$$

Indeed, if all the variances $\Sigma_1, \ldots, \Sigma_n$ were equal, then the error would be a multiple of the square of the length of the vector $\boldsymbol{b} - A\boldsymbol{v}$, so we would be solving the ordinary least squares problem. But if the Σ_i's are different, this means that we weigh the error by the inverse of the variance: those measurements b_i with smaller (respectively larger) variance are more (respectively less) reliable, hence they will weight more (respectively less) in the error. Still, the problem of minimizing (302) is actually equivalent with the ordinary least squares problem for the related system:

$$A'\boldsymbol{v} \sim \boldsymbol{b}', \qquad \text{where} \quad A' = D^{-\frac{1}{2}}A, \quad \boldsymbol{b}' = D^{-\frac{1}{2}}\boldsymbol{b} \quad \text{and} \quad D = \begin{bmatrix} \Sigma_1 & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \Sigma_n \end{bmatrix}$$

Therefore, the least squares solution reads:

$$\boldsymbol{v} = (A'^T A')^{-1} A'^T \boldsymbol{b}' = (A^T D^{-1} A)^{-1} A^T D^{-1} \boldsymbol{b} \tag{303}$$

In terms of the system (301), this is known as **weighted least squares**. The most general scenario is when the random variables $b_1, \ldots, b_n$ are not independent, and they are related by the covariance matrix K. In this case, the appropriate analogue of the solution (303) is:

$$\boldsymbol{v} = (A^T K^{-1} A)^{-1} A^T K^{-1} \boldsymbol{b} \tag{304}$$

and it is known as **generalized least squares**. It is precisely the choice of $\boldsymbol{v}$ which minimizes the error, whose appropriate definition is:

$$\varepsilon = (\boldsymbol{b} - A\boldsymbol{v})^T K^{-1} (\boldsymbol{b} - A\boldsymbol{v})$$

Another question you might ask is what is the covariance matrix of the solution (304) in terms of the covariance matrix $K = E[\boldsymbol{b}\boldsymbol{b}^T]$. Assume that the random vector $\boldsymbol{b}$ has zero mean, to keep notations simple (it is reasonable to expect that measuring errors will have zero mean). Then:

$$E[\boldsymbol{v}\boldsymbol{v}^T] = (A^T K^{-1} A)^{-1} A^T K^{-1} \underbrace{E[\boldsymbol{b}\boldsymbol{b}^T]}_{K} K^{-1,T} A (A^T K^{-1,T} A)^{-1} =$$

$$= (A^T K^{-1} A)^{-1} A^T K^{-1,T} A (A^T K^{-1,T} A)^{-1} = (A^T K^{-1} A)^{-1}$$

I hope this has been a convincing introduction (although we have just scratched the surface) of how linear algebra is useful in probability.